

TruthLens AI

Technical Documentation

1. Project Title

TruthLens AI – AI-Powered Misinformation & Credibility Verification Platform

2. Team Information

Team Name: TruthLens AI

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Institution: CMR University, Bengaluru

3. Problem Statement

The rapid growth of digital media has resulted in the widespread distribution of misinformation, fake news, and manipulated content across social media and online platforms.

Users often struggle to determine whether information is trustworthy before sharing it. Existing solutions are either complex, slow, or lack transparency in how credibility scores are generated.

TruthLens AI addresses this challenge by providing an explainable and user-friendly credibility verification system.

4. Proposed Solution

TruthLens AI is a web-based platform that analyzes textual content and source information to generate a credibility assessment report.

The system:

- Analyzes user-provided content
- Evaluates source credibility

- Calculates a trust score
- Determines risk level
- Provides transparent explanations
- Generates downloadable PDF verification reports

The platform follows the ASEEV Framework:

ASEEV Framework

A – Analyze Content

Identifies suspicious patterns and keywords.

S – Score Credibility

Computes overall trust score.

E – Evaluate Risk

Classifies content into risk categories.

E – Explain Findings

Provides transparent explanations.

V – Verify Information

Generates verification report for users.

5. Objectives

The main objectives of TruthLens AI are:

- Detect potentially misleading content
 - Improve media literacy
 - Encourage responsible content sharing
 - Provide explainable AI-driven verification
 - Generate downloadable credibility reports
 - Increase trust in digital information
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6. System Architecture



7. Technology Stack

Frontend

- React.js
- Vite
- Axios

Purpose

Provides an interactive user interface for content analysis and report generation.

Backend

- FastAPI
- Python

Purpose

Processes requests and executes the ASEEV framework logic.

Libraries Used

Axios

Used for communication between frontend and backend.

jsPDF

Used for generating downloadable PDF verification reports.

Pydantic

Used for request validation in FastAPI.

8. Functional Modules

Module 1: Content Analysis

Analyzes text entered by users.

Detects:

- Breaking News phrases
 - Sensational language
 - Viral propagation indicators
 - Secret or unverified claims
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Module 2: Source Credibility Assessment

Evaluates source URLs.

Examples:

High Credibility

- BBC
- Reuters
- The Hindu

Low Credibility

- Fake domains
 - Unknown xyz domains
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Module 3: Credibility Scoring Engine

Initial trust score starts at:

95

Score is reduced based on:

Indicator	Penalty
Breaking News Phrase	-25
Shocking Language	-25
Viral Content Indicator	-20
Unverified Secret Claim	-20
Low Credibility Source	-20

Module 4: Risk Assessment

Low Risk

Trust Score $\geq$ 80

Medium Risk

Trust Score 60–79

High Risk

Trust Score < 60

Module 5: Explainable AI Output

Provides transparent explanations such as:

- Breaking News Phrase Detected
- Sensational Language Detected

- Viral Propagation Pattern Detected
- Unverified Secret Claim Detected

Module 6: PDF Report Generation

Users can download a verification report containing:

- Trust Score
- Risk Level
- AI Confidence
- Source Credibility
- Source URL
- Analysis Reasons
- Timestamp

9. Workflow

User Enters Content



Analyze Content



Score Credibility



Evaluate Risk



Explain Findings



Verify Information



Generate PDF Report

10. Key Features

- Real-time content verification
 - Trust score generation
 - Risk assessment
 - Source credibility analysis
 - Explainable AI outputs
 - PDF report generation
 - User-friendly interface
 - ASEEV framework implementation
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11. Future Enhancements

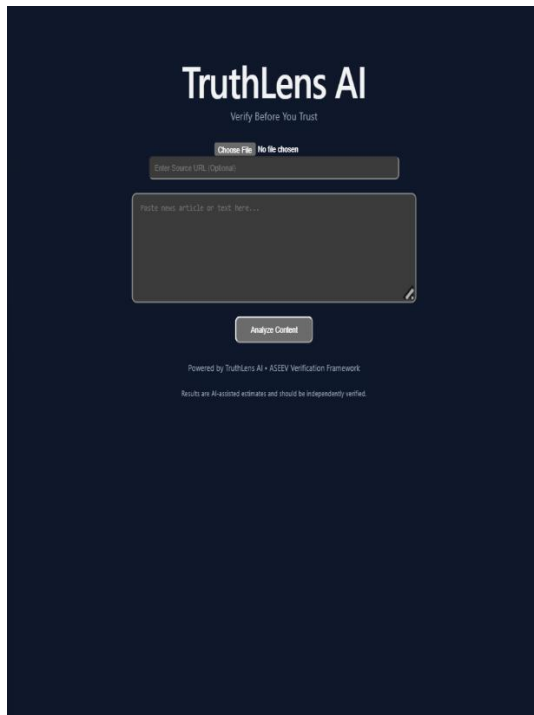
Future versions of TruthLens AI may include:

- Transformer-based NLP models
 - Deepfake image detection
 - Fact-checking APIs
 - Multilingual verification
 - Source reputation database
 - Browser extension integration
 - Social media verification plugin
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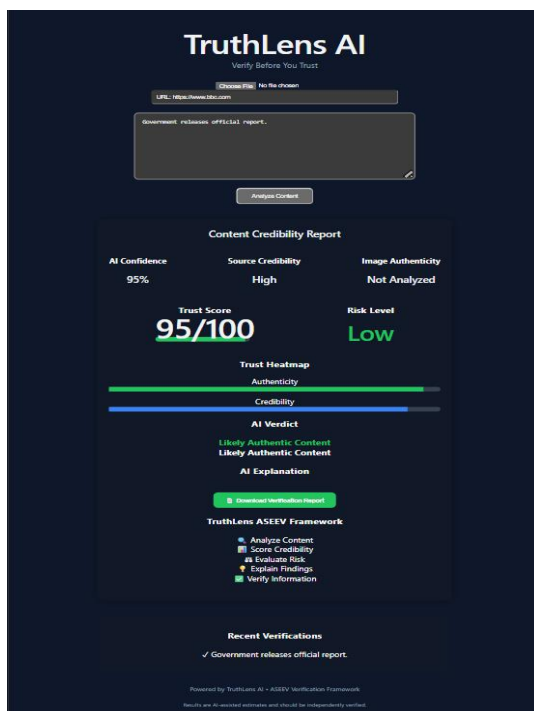
12. Sample Output Screenshots

The following screenshots demonstrate the functionality of TruthLens AI:

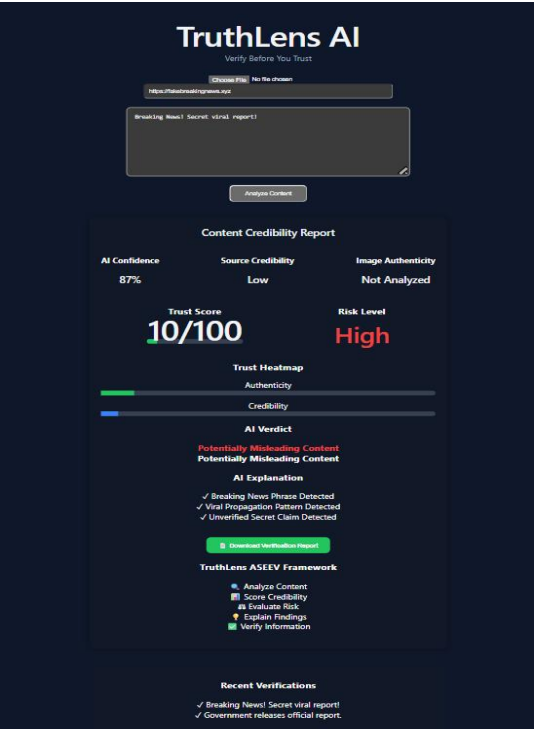
- Homepage Screenshot



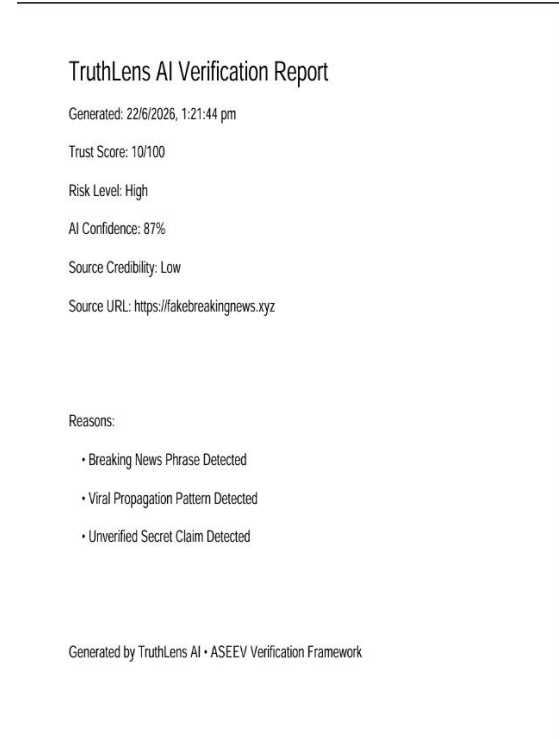
- High Credibility Analysis (BBC)



- Low Credibility Analysis (Fake URL)



- PDF Verification Report



13. Business Impact

TruthLens AI helps:

- Reduce misinformation spread
- Promote responsible content sharing
- Improve media literacy
- Support educational institutions
- Assist journalists and researchers
- Increase transparency in AI-driven verification

14. Conclusion

TruthLens AI provides a practical, explainable, and scalable approach to misinformation detection through its ASEEV framework. By combining content analysis, source credibility assessment, risk evaluation, and downloadable verification reports, the platform empowers users to make informed decisions before trusting or sharing information.

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Project: TruthLens AI

Framework: ASEEV (Analyze, Score, Evaluate, Explain, Verify)

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